Trade Study: LiDAR Options

- **Conducted by:** Travis Ohm and Alex Bloemker
- **Decision:** RPLiDAR C1M1-R2

Problem: *A LiDAR sensor is required because the project's core goal, which is detecting, tracking, and following an object, needs accurate, real-time distance data that cameras alone can't reliably provide in all lighting conditions. LiDAR delivers direct range measurements independent of ambient light or object color, which is critical for consistent tracking and safe navigation. It also sets up the platform for future obstacle-avoidance and mapping capabilities as the sensor-fusion architecture expands.*

		RPLiDAR C1M1-R2	*DFRobot TFS20-L*	*DFRobot SEN0646*
Criteria	**Weights**	**Grade**	**Grade**	**Grade**
Range	0.15	7	9	4
Accuracy	0.15	7	6	8
Update Rate	0.10	6	10	6
FOV/ Coverage	0.20	10	3	3
Jetson Integration	0.15	9	5	5
Power Consumption	0.10	5	9	8
Cost	0.10	5	7	10
Weight/ Size	0.05	4	10	10
Average Score		6.63	7.38	6.75
Weighted Score		7.25	6.70	6.05

Breakdown

1st Choice: RPLiDAR C1M1-R2

Only candidate with 360° coverage, which is a decisive advantage for object tracking and future obstacle-avoidance/mapping needs, which is why Field of View/Coverage carries the highest weight (0.20), and this option scores a 10 there. It has native ROS/ROS2 SDK support and an active user base substantially de-risks integration within a tight sprint. The Trade-offs are the highest cost ($69), heaviest (110g), and highest power draw (~1.15W) of the three, which is acceptable given the coverage and integration wins.

2nd Choice: DFRobot TFS20-L

It has the best raw range (20m) and by far the fastest update rate (up to 250Hz), plus strong outdoor/sunlight robustness (100 Klux) and low power draw (< 0.43 W). It is mainly held back by its single-point (<2° FoV) sensing, meaning it can't provide the coverage needed for tracking a moving object across a field of view, only fixed-direction distance. It has no ROS-native support, just Arduino/ESP32/RPi libraries, adding integration effort on the Jetson/ROS side.

3rd Choice: DFRobot SEN0646

It is the cheapest ($15) and smallest/lightest (1g) by a wide margin, with the best accuracy (±4cm) of the three at short range. It contains the same single-point coverage limitation as the TFS20-L, plus the shortest max range (7.8m) and a lower update rate (50Hz) — the weakest overall fit for the primary tracking sensor role. It is best suited as a low-cost supplementary sensor (e.g., close-range collision detection) rather than the main ranging solution.